

Title Placeholder

Anonymous submission

Abstract

Time-series anomaly detection aims to identify abnormal segments that deviate from normal temporal behaviors. Recent multimodal methods have shown the potential of incorporating textual information for anomaly detection. However, external textual sources, such as logs, events, or manual descriptions, are often noisy, weakly aligned with numerical signals, and mixed with task-irrelevant semantics, which may interfere with reconstruction-based anomaly scoring. To address this issue, we propose a text-temporal multimodal reconstruction framework for time-series anomaly detection. Instead of relying on external textual annotations or directly asking language models to judge anomalies, our framework derives textual descriptions from the time series itself to summarize segment-level temporal behaviors. The descriptions are encoded by a frozen large language model and aligned with numerical patch representations through a probabilistic text-temporal alignment objective. The aligned textual features are then incorporated into a cross-modal masked reconstruction module to guide temporal patch recovery. During inference, reconstruction errors are used as anomaly scores. Experiments on benchmark datasets demonstrate that our framework consistently improves anomaly detection performance.

Introduction

Time-series anomaly detection (TSAD) aims to identify observations or segments that deviate from expected temporal behaviors, and has been widely studied in anomaly detection research (??). It plays a critical role in real-world monitoring scenarios such as spacecraft telemetry, industrial systems, cloud services, cyber-physical systems, and large-scale sensor networks (??). Since abnormal events are usually rare, diverse, and expensive to annotate, unsupervised reconstruction-based methods have become a widely adopted paradigm. Representative studies learn normal temporal patterns through density estimation, reconstruction, or hierarchical representation learning, such as DAGMM (?), MSCRED (?), and THOC (?). These methods show that modeling regular temporal dynamics from historical observations is effective for detecting abnormal deviations.

This is an anonymized submission for review purposes only. Distribution, citation, or public sharing of this manuscript is strictly prohibited. Copyright and publication details will appear in the final version if accepted.

Despite their progress, most reconstruction-based TSAD methods still operate mainly in the numerical space. They recover raw values or latent representations from corrupted inputs, while the high-level behaviors behind a time-series window are usually learned only implicitly. In practice, anomalies may not only appear as point-wise deviations, but also as abnormal local variation, trend evolution, periodic dependency, lag dependency, or component-level changes. Prior time-series modeling studies have shown that decomposition, periodicity, frequency-domain structures, and temporal dependencies are essential for understanding sequential data (???). However, these temporal structures are rarely transformed into explicit textual representations for reconstruction-based anomaly detection. As a result, existing methods may overlook useful behavioral cues that are difficult to capture from numerical reconstruction alone.

Recent advances in foundation models, language models, and multimodal learning provide new opportunities for introducing textual information into time-series modeling. TimeGPT, MOMENT, and Lag-Llama demonstrate the potential of large-scale pretraining for general time-series analysis (???), while recent studies also begin to investigate whether language models can understand time-series anomalies (?). However, existing attempts to incorporate textual information still face several limitations. Directly serializing raw numerical values into text usually produces lengthy inputs with limited behavioral abstraction, while methods relying on external textual information may suffer from noise, redundancy, and weak alignment with the target time-series window. More importantly, textual information is often used as auxiliary prompts or decision-level cues, which is insufficient for reconstruction-based TSAD where anomaly scoring depends on temporal recovery. Therefore, textual information should be derived from the time series itself and transformed into reconstruction-compatible signals that can interact with numerical patch representations.

To bridge the gap between textual temporal information and numerical temporal recovery, we propose a text-temporal reconstruction framework for time-series anomaly detection. Given an input window, we divide the time series into local patches and construct dual-view textual information directly from the time series itself. The behavior description view summarizes segment-level temporal patterns, while the distribution prediction view transforms textual information into

reconstruction-oriented predictive distributions. These text-derived distributions are then aligned with numerical patch representations to guide masked temporal reconstruction. In this way, our method converts time-series-derived textual information into reconstruction-compatible signals, allowing textual information to directly contribute to reconstruction-based anomaly scoring.

Our main contributions are summarized as follows:

- We propose a text-temporal reconstruction framework for TSAD that incorporates time-series-derived textual information into masked temporal reconstruction, moving beyond purely numerical patch recovery.
- We design a dual-view text construction strategy from the input time-series window, where the behavior description view summarizes segment-level temporal patterns and the distribution prediction view converts textual information into reconstruction-oriented predictive distributions.
- We develop a distribution-guided fusion mechanism that aligns text-derived predictive distributions with numerical patch representations, enabling textual information to guide masked patch recovery and improve reconstruction-based anomaly scoring.

Related Work

Time-Series Anomaly Detection

Time-series anomaly detection (TSAD) aims to identify observations or segments that deviate from normal temporal patterns, and has been widely studied in industrial monitoring, cloud systems, cyber-physical systems, and sensor networks. Deep TSAD methods commonly follow reconstruction-, prediction-, density-estimation-, or representation-learning-based paradigms. OmniAnomaly models multivariate temporal dynamics with stochastic recurrent networks and detects anomalies using reconstruction probability (?), while USAD introduces adversarially trained autoencoders for efficient unsupervised anomaly detection (?). To better capture dependencies among variables, MTAD-GAT combines forecasting and reconstruction with temporal and feature-oriented graph attention (?), and GDN learns sensor relationships to detect deviations from graph-based normal behavior (?). InterFusion further models inter-metric and temporal dependencies through hierarchical latent variables (?). More recently, Transformer-based methods have improved long-range dependency modeling. Anomaly Transformer proposes association discrepancy to distinguish abnormal temporal associations (?), TranAD adopts self-conditioning and adversarial training for multivariate anomaly detection (?), and DCdetector explores dual-attention contrastive representation learning for anomaly discrimination (?).

Another relevant line of work is patch-based and masked time-series modeling. PatchTST represents time series as local patches, reducing attention cost while preserving local temporal semantics (?). TimesNet transforms one-dimensional series into two-dimensional temporal variations to capture multi-periodic patterns (?). Ti-MAE and SimMTM show that masked reconstruction can encourage

models to learn informative temporal representations from corrupted time-series inputs (?). These studies suggest that local patches, temporal variations, periodic structures, and masked recovery are important for time-series representation learning. However, most existing TSAD methods still operate mainly in the numerical space. Segment-level behaviors such as local variation, trend evolution, periodic dependency, lag dependency, and component-level changes are usually learned implicitly by neural networks rather than explicitly represented. This motivates our work to construct textual behavioral descriptions from the time series itself and use them to assist reconstruction-based anomaly detection.

Multimodal Time-Series Analysis

Recent studies have explored modeling time series beyond purely numerical representations by introducing language models and multimodal learning. Some methods reformulate time-series forecasting into language-compatible tasks, such as sentence-to-sentence prediction or numerical value serialization (?). Others tokenize time series or adapt pre-trained foundation models through alignment and reprogramming, enabling language-model architectures to process temporal data (????). In addition, decomposition-aware prompting and temporal structure modeling have been introduced to better adapt language models to time-series analysis (?). These studies show the potential of language-model priors for temporal modeling, but they mainly focus on forecasting or general representation learning rather than reconstruction-based anomaly detection.

Multimodal time-series analysis further investigates how numerical signals interact with auxiliary modalities such as text, images, events, and domain knowledge (???). For TSAD, recent works convert time series into textual or visual forms for LLM/MLLM-based anomaly detection, or explore time-text alignment for multimodal anomaly modeling (???). However, these methods often rely on raw-value serialization, visualized inputs, external textual cues, or decision-level reasoning. In contrast, our method derives textual descriptions directly from the input time-series window and transforms them into predictive distributions aligned with numerical patch representations, allowing textual temporal information to participate in masked reconstruction and anomaly scoring.

Methodology

Problem Statement

Given an input window $\mathbf{X} \in \mathbb{R}^{L \times C}$, where L denotes the window length and C represents the number of variables, the model learns a reconstruction mapping $\hat{\mathbf{X}} = f_{\theta}(\mathbf{X})$. During inference, the reconstruction error serves as the anomaly score. Let $x_{t,c}$ and $\hat{x}_{t,c}$ denote the c -th variable at timestamp t of the input and its reconstruction, respectively. The anomaly score is computed as the channel-averaged reconstruction error:

$$s_t = \frac{1}{C} \sum_{c=1}^C (x_{t,c} - \hat{x}_{t,c})^2. \quad (1)$$

A larger s_t indicates that the observation is harder to reconstruct under the learned normal patterns, and is therefore

Figure 1: Overview of Ours Framework

Left: Input window normalization, patching, and masking
Middle: Dual-view textual modality construction (Statistical prompt view + Structural component view)
Right: Text-temporal alignment and masked reconstruction

Figure 1: Overview of the proposed Ours framework for multi-view time series anomaly detection with textual fusion.

more likely to be anomalous.

Overview

The proposed framework reconstructs each input window by jointly modeling numerical temporal patterns and structured textual descriptions derived from the time series itself. As illustrated in Figure 1, Ours consists of three main stages: (1) temporal patch encoding with masking, (2) dual-view textual modality construction, and (3) text-temporal alignment for masked reconstruction.

Given an input window $\mathbf{X} \in \mathbb{R}^{L \times C}$, we first apply window-level normalization:

$$\bar{\mathbf{X}} = \frac{\mathbf{X} - \mathbf{m}}{\sigma}, \quad \sigma = \sqrt{\text{Var}(\mathbf{X}) + \epsilon}, \quad (2)$$

where $\mathbf{m}$ and σ are computed along the temporal dimension and ϵ is a small constant for numerical stability. The normalized sequence is divided into patches with size p and stride r , yielding $N_p = \lfloor (L - p)/r \rfloor + 1$ patches. Each patch is projected to dimension d via a learnable linear projection $\mathbf{W}_{\text{patch}} \in \mathbb{R}^{p \times d}$, followed by additive positional encoding. The resulting patch representation is $\mathbf{P} \in \mathbb{R}^{B \times C \times N_p \times d}$, where B is the batch size and d is the hidden dimension. A random patch mask is applied to construct incomplete temporal evidence. Both the complete and masked patch sequences are encoded by a shared Transformer encoder E_t , where $\text{Mask}(\cdot)$ randomly zeros out a fraction of patch tokens:

$$\mathbf{H}^t = E_t(\mathbf{P}), \quad \tilde{\mathbf{H}}^t = E_t(\text{Mask}(\mathbf{P})), \quad (3)$$

where $\mathbf{H}^t$ serves as the complete temporal representation for alignment and $\tilde{\mathbf{H}}^t$ serves as the masked temporal representation for reconstruction.

Textual Modality Construction

The textual modality is constructed from two complementary views of the input window.

Statistical prompt view. For each channel-wise patch, the model extracts statistical descriptors including the minimum, maximum, and median values, and dominant lags computed via autocorrelation. It also estimates a coarse window-level trend direction from first-order differences across patches and inserts this trend descriptor into each patch prompt. These descriptors are formatted with dataset context into a natural-language prompt and encoded by a frozen large language model. Let $\mathbf{H}^{\text{llm}} \in \mathbb{R}^{B \times C \times N_p \times M \times d_{\text{llm}}}$ denote the LLM hidden states, where M is the maximum prompt length. These are first projected to the temporal latent space via $\mathbf{W}_h \in \mathbb{R}^{d_{\text{llm}} \times d}$, flattened over the token dimension, and then projected to dimension d via $\mathbf{W}_{\text{prompt}} \in \mathbb{R}^{(M \cdot d) \times d}$:

$$\begin{aligned} \mathbf{U}' &= \text{flatten}(\Phi(\bar{\mathbf{X}})\mathbf{W}_h) \mathbf{W}_{\text{prompt}}, \\ \mathbf{U} &= \text{reshape}_{B \times C \times N_p \times d}(\mathbf{U}'). \end{aligned} \quad (4)$$

where $\Phi(\cdot)$ denotes the prompt construction function that returns $\mathbf{H}^{\text{llm}}$, and $\text{flatten}(\cdot)$ reshapes $\mathbb{R}^{B \times C \times N_p \times M \times d}$ to $\mathbb{R}^{B \times C \times N_p \times (M \cdot d)}$. Accordingly, $\mathbf{U} \in \mathbb{R}^{B \times C \times N_p \times d}$. Since the LLM parameters are frozen, the language model serves as a text encoder rather than a trainable anomaly detector.

Structural component view. The raw input window is decomposed into trend, seasonal, and residual components:

$$\mathbf{X} = \mathbf{X}^{tr} + \mathbf{X}^{se} + \mathbf{X}^{re}. \quad (5)$$

The trend component is obtained by a moving-average filter followed by a linear temporal mapping; the seasonal component is estimated from the detrended signal via a feed-forward network; and the residual component is the remaining signal. Each component is independently normalized and patch-embedded. Let $\mathbf{R}^q \in \mathbb{R}^{M \times d}$ denote the LLM-encoded role prompt for component $q \in \{tr, se, re\}$. The component patch embeddings are concatenated with their respective role prompts and encoded by a component encoder E_c shared across components:

$$\mathbf{V}^q = \Pi_{\text{patch}}(E_c([\mathbf{R}^q; \mathbf{P}^q])), \quad (6)$$

where $[\cdot; \cdot]$ denotes concatenation along the patch dimension, and $\Pi_{\text{patch}}(\cdot)$ selects the N_p output tokens corresponding to component patches after removing the role prompt tokens. The encoded representations of the trend, seasonal, and residual components are denoted as $\mathbf{V}^{tr}$, $\mathbf{V}^{se}$, and $\mathbf{V}^{re}$, respectively. These are fused via a learnable linear projection $\mathbf{W}_c \in \mathbb{R}^{3d \times d}$:

$$\mathbf{V} = [\mathbf{V}^{tr}; \mathbf{V}^{se}; \mathbf{V}^{re}] \mathbf{W}_c, \quad (7)$$

where $[\cdot; \cdot; \cdot]$ denotes concatenation along the feature dimension. Accordingly, $\mathbf{U}$ captures prompt-level statistical features and $\mathbf{V}$ captures structure-aware temporal features.

Text-Temporal Alignment

The two textual views are integrated via a cross-attention layer followed by a feed-forward network. Denoting multi-head attention as $\mathcal{A}(Q, K, V)$, layer normalization as $\text{LN}(\cdot)$, and a feed-forward network as $\text{FFN}(\cdot)$, the fusion is formulated as a standard Transformer block:

$$\mathbf{Z}_1 = \text{LN}(\mathbf{U} + \mathcal{A}(\mathbf{U}, \mathbf{V}, \mathbf{V})), \quad (8)$$

$$\mathbf{Z} = \text{LN}(\mathbf{Z}_1 + \text{FFN}(\mathbf{Z}_1)), \quad (9)$$

where $\mathbf{U}$ serves as the query and $\mathbf{V}$ serves as the key and value. The output $\mathbf{Z} \in \mathbb{R}^{BC \times N_p \times d}$ is the fused textual representation of the input window.

To align the textual representation with the temporal representation, we model the conditional distribution of the temporal patches given the textual representation as a diagonal Gaussian distribution. The textual representation predicts the mean and log-variance tensors:

$$\boldsymbol{\mu} = f_\mu(\mathbf{Z}), \quad \boldsymbol{\ell} = \text{clip}(f_\sigma(\mathbf{Z})). \quad (10)$$

where f_μ and f_σ are linear projections and $\ell = \log \sigma^2$. For each training window, let $c \in \{1, \dots, C\}$ index the channel dimension, $i \in \{1, \dots, N_p\}$ index the patch dimension, and $j \in \{1, \dots, d\}$ index the feature dimension. The conditional distribution of each temporal patch is

$$p(\mathbf{h}_{c,i}^t | \mathbf{z}_{c,i}) = \mathcal{N}(\mathbf{h}_{c,i}^t; \boldsymbol{\mu}_{c,i}, \text{diag}(\exp(\boldsymbol{\ell}_{c,i}))). \quad (11)$$

The alignment objective minimizes the Gaussian negative log-likelihood up to an additive constant:

$$\mathcal{L}_{\text{align}} = \mathbb{E}_{\mathbf{X} \sim \mathcal{D}} \left[\frac{1}{2CN_p d} \sum_{c=1}^C \sum_{i=1}^{N_p} \sum_{j=1}^d \left(\ell_{c,i,j} + (\text{sg}(h_{c,i,j}^t) - \mu_{c,i,j})^2 \exp(-\ell_{c,i,j}) \right) \right]. \quad (12)$$

where $\text{sg}(\cdot)$ denotes the stop-gradient operator that prevents gradients from flowing into the temporal encoder. The operator $\text{clip}(\cdot)$ constrains each log-variance element to a stable range. This probabilistic formulation encourages the textual representation to predict both the location and the uncertainty of the temporal representation, thereby accommodating the inherent variability in text-temporal correspondences.

Reconstruction Module

The reconstruction module leverages the textual representation to guide the recovery of masked temporal features. The masked temporal features first undergo self-attention to aggregate partial temporal context:

$$\tilde{\mathbf{H}}^t = \mathcal{A}(\tilde{\mathbf{H}}^t, \tilde{\mathbf{H}}^t, \tilde{\mathbf{H}}^t). \quad (13)$$

The textual representation then queries this self-attended temporal memory via cross-attention, followed by a feed-forward network:

$$\mathbf{G}_1 = \text{LN}(\mathbf{Z} + \mathcal{A}(\mathbf{Z}, \tilde{\mathbf{H}}^t, \tilde{\mathbf{H}}^t)), \quad (14)$$

$$\mathbf{G} = \text{LN}(\mathbf{G}_1 + \text{FFN}(\mathbf{G}_1)), \quad (15)$$

where $\tilde{\mathbf{H}}^t$ serves as the key and value, and $\mathbf{Z}$ serves as the query. This design enables reconstruction to benefit from both textual regularities and partial temporal evidence.

Finally, $\mathbf{G}$ is flattened over patches and projected back to the input space:

$$\hat{\mathbf{X}} = \text{reshape}_{B \times L \times C}(\text{flatten}(\mathbf{G}) \mathbf{W}_o), \quad (16)$$

where $\mathbf{W}_o \in \mathbb{R}^{(N_p \cdot d) \times L}$ is a learnable output projection, and $\text{flatten}(\cdot)$ reshapes $\mathbf{G} \in \mathbb{R}^{BC \times N_p \times d}$ to $\mathbb{R}^{BC \times (N_p \cdot d)}$. After inverse normalization, the reconstruction is compared with the original input for anomaly scoring.

Training Objective

The reconstruction loss is the mean squared error between the input and its reconstruction:

$$\mathcal{L}_{\text{rec}} = \mathbb{E}_{\mathbf{X} \sim \mathcal{D}} \left[\frac{1}{LC} \left\| \mathbf{X} - \hat{\mathbf{X}} \right\|_F^2 \right], \quad (17)$$

where $\mathcal{D}$ denotes the training-window distribution and $\|\cdot\|_F$ denotes the Frobenius norm over the temporal and channel dimensions. The alignment loss $\mathcal{L}_{\text{align}}$ is the Gaussian negative log-likelihood defined in the previous subsection. The total loss combines reconstruction and alignment:

$$\mathcal{L} = \mathcal{L}_{\text{rec}} + \lambda \mathcal{L}_{\text{align}}, \quad (18)$$

where λ is the weighting coefficient for alignment loss. By jointly optimizing these terms, the model learns normal temporal reconstruction and probabilistic text-temporal alignment. During inference, timestamps with larger reconstruction errors are assigned higher anomaly scores.

Experiments

Experimental Setup

Datasets. We evaluate Ours on six real-world datasets: *Weather* (meteorological measurements), *Energy* (energy prices and statistics), *Environment* (air quality), and the financial series *KR*, *EWJ*, and *MDT*. Weather, Energy, and Environment are collected from Time-MMD (?), whereas the financial series are derived from FNSPID (?). KR, EWJ, MDT, and Environment are univariate; Energy and Weather contain 9 and 4 variables, respectively. We split every series chronologically into 64% training, 16% validation, and 20%

Table 1: Main results on six benchmark datasets. Aff-F is reported under the mean-threshold protocol, while V-PR and V-ROC are threshold-independent. Values are percentages. Bold = best; underline = second best.

Dataset	Metric	Ours	MindTS	DLinear	PatchTST	iTrans.	Cross.	LightTS	NLinear	Linear	TimesNet	FiLM	Informer	Reformer	Trans.	MICN	FEDformer
Weather	Aff-F	72.33	<u>69.86</u>	47.72	50.32	49.84	43.03	38.61	39.23	37.97	50.20	39.78	40.01	36.36	35.39	39.82	27.75
	V-PR	48.53	<u>47.91</u>	45.37	47.18	<u>48.46</u>	44.63	42.15	39.46	42.81	47.87	46.43	47.09	46.52	47.03	42.29	34.73
	V-ROC	77.07	76.41	74.28	76.13	<u>77.00</u>	73.62	71.17	71.78	72.88	76.08	71.37	76.22	75.49	76.11	69.96	52.52
Energy	Aff-F	65.22	<u>63.10</u>	48.92	45.29	44.53	37.94	45.57	45.23	41.13	47.65	42.56	39.81	36.90	35.63	44.64	35.64
	V-PR	45.83	<u>41.10</u>	<u>44.73</u>	40.80	36.62	39.87	34.99	31.77	31.66	41.99	39.60	35.33	35.09	34.82	37.88	32.40
	V-ROC	71.96	67.16	<u>71.43</u>	68.48	65.36	68.39	63.82	58.87	56.70	67.71	63.87	58.51	57.79	58.24	63.53	54.21
Environment	Aff-F	65.21	<u>63.46</u>	37.19	36.54	31.17	26.99	31.52	31.13	30.76	35.84	22.93	36.37	37.87	43.55	25.58	31.15
	V-PR	39.69	36.91	37.48	38.36	<u>39.23</u>	22.76	31.25	36.45	34.49	21.76	19.51	10.17	14.51	10.90	10.24	7.98
	V-ROC	86.81	83.49	85.13	82.28	<u>85.47</u>	72.48	81.59	84.97	84.31	65.73	72.49	51.26	59.98	55.56	48.45	42.21
KR	Aff-F	79.31	<u>77.03</u>	59.02	61.54	55.88	57.32	56.05	52.30	49.72	50.39	56.04	46.91	44.22	42.34	49.88	40.26
	V-PR	55.53	<u>54.82</u>	54.37	47.05	29.71	36.03	42.26	34.70	38.73	19.75	38.06	21.60	14.44	14.33	19.63	16.23
	V-ROC	93.29	<u>90.53</u>	<u>91.13</u>	87.96	75.43	73.96	85.70	78.04	83.87	69.74	80.34	54.52	62.49	55.45	59.92	65.02
EWJ	Aff-F	72.67	<u>62.69</u>	56.01	48.12	50.81	49.58	51.00	45.63	51.04	49.83	31.21	46.30	39.69	43.06	38.01	33.36
	V-PR	48.52	25.60	<u>47.36</u>	37.01	37.87	29.65	35.84	33.56	30.25	31.46	15.28	30.52	20.64	19.37	15.99	16.41
	V-ROC	82.70	65.33	<u>81.08</u>	74.82	76.56	68.49	73.52	74.39	71.20	63.33	51.91	73.29	53.70	55.73	50.85	56.00
MDT	Aff-F	69.25	<u>64.72</u>	47.95	48.13	41.17	42.26	38.60	37.18	36.49	36.08	39.78	38.76	36.39	42.56	38.86	36.51
	V-PR	59.14	48.31	<u>53.44</u>	51.46	38.62	32.47	24.92	27.96	21.41	28.01	40.16	23.02	16.91	19.47	27.13	19.14
	V-ROC	81.65	75.33	<u>77.17</u>	<u>78.14</u>	72.56	62.57	59.20	61.17	50.79	65.82	70.16	51.75	41.19	47.60	57.50	51.18

test data: the benchmark first reserves the last 20% for testing and then uses the last 20% of the remaining prefix for validation. The released labels confirm that both training and validation portions are anomaly-free; anomalies occur only in the test portion. Channel-wise standardization statistics are fitted on the training portion and then applied unchanged to validation and test data. Training and validation windows are extracted with a one-timestamp step, while test windows are tiled without overlap to produce one score per timestamp.

Baselines. We compare with Linear, DLinear, and NLinear (?); PatchTST (?), TimesNet (?), MICN (?), FiLM (?), and LightTS (?); Transformer (?), iTransformer (?), Crossformer (?), Informer (?), Reformer (?), and FEDformer (?); and MindTS (?), the closest text-temporal baseline. Forecasting backbones are converted into detectors through their reconstruction/anomaly-detection branches rather than by forecasting future samples. Each receives a window $\mathbf{X} \in \mathbb{R}^{L \times C}$ and outputs an equal-length $\hat{\mathbf{X}} \in \mathbb{R}^{L \times C}$ (forecast horizon zero). In particular, DLinear reconstructs the window through its seasonal-trend linear head, PatchTST uses a length- L patch-reconstruction head, iTransformer projects its variable-token representation back to L timestamps, and Informer-family models use their encoder-side reconstruction projection. The remaining forecasting backbones are adapted through the corresponding full-window anomaly-detection branch in the same library. All models minimize full-window reconstruction MSE; Ours additionally uses its alignment loss. Because the comparison does not include a broad set of dedicated TSAD methods, we interpret it as a controlled comparison among the evaluated reconstruction backbones, not as a comprehensive claim of TSAD state of the art.

Evaluation Metrics. We adopt Affiliated-F1-score (Aff-F) (?) as the primary metric for segment-level detection quality and report it using the benchmark’s mean-threshold protocol. We also report V-PR and V-ROC (?), which evaluate range-aware detection quality over threshold and window variations.

Implementation Details. Ours is implemented in PyTorch. Qwen3-1.7B is frozen and truncated to six Transformer layers; both the temporal encoder and text-guided reconstruction branch contain one Transformer layer. We use

Adam for 5 epochs with an initial learning rate of 10^{-4} and the type-1 step schedule, which halves the rate at successive epoch updates. The alignment term is Gaussian NLL with $\lambda = 1.0$. The window/patch/patch-stride settings ($L/p/r$) are 24/8/8 for KR, 48/6/6 for EWJ, 72/6/6 for Environment, and 24/6/6 for MDT, Energy, and Weather. The mask ratio is 0.3 on EWJ and 0.4 otherwise. Batch sizes are 16 for KR and EWJ, 32 for MDT and Energy, and 64 for Environment and Weather. We report one run with seed 2021, matching the benchmark configuration, on a server with two NVIDIA RTX 4090 GPUs (24 GB each).

Main Results

Table 1 shows that Ours ranks first in all 18 dataset-metric combinations. The appropriate strongest baseline differs by metric. For Aff-F, Ours averages 70.67%, compared with 66.81% for MindTS, a gain of 3.86 percentage points. For V-PR, the strongest baseline is DLinear at 47.13%; Ours reaches 49.54%, improving by 2.42 points. DLinear is also strongest in average V-ROC at 80.04%, whereas Ours obtains 82.25%, a gain of 2.21 points. The aggregate advantage therefore holds for both the mean-threshold segment metric and the two threshold-independent range-aware metrics.

MindTS remains the closest comparison in model setting because it also combines text and time series. Relative to MindTS, Ours improves the three averages from 66.81% to 70.67% in Aff-F, from 42.44% to 49.54% in V-PR, and from 76.38% to 82.25% in V-ROC, corresponding to gains of 3.86, 7.10, and 5.87 points. At the dataset level, the largest Aff-F margin is on EWJ (9.98 points), while the margins on Energy, KR, and MDT are 2.12, 2.28, and 4.53 points. These results describe consistent empirical improvements across the evaluated datasets; the contribution of individual mechanisms is examined separately through ablation.

LLM Backbone Analysis

Table 2 evaluates seven lightweight backbones with 1.0B–1.7B parameters on all three metrics. Qwen3-1.7B gives the best average Aff-F (70.67%) and V-PR (49.54%). OLMo-1B gives the highest average V-ROC (82.57%), slightly above Qwen3-1.7B (82.25%) by 0.33 points. Thus, no single backbone is best for every evaluation criterion.

Table 2: LLM backbone comparison. Aff-F is reported under the mean-threshold protocol; V-PR and V-ROC are threshold-independent. Values are percentages. Bold = best; underline = second best.

Model	Weather			Energy			Env.			KR			EWJ			MDT		
	Aff-F	V-PR	V-ROC	Aff-F	V-PR	V-ROC	Aff-F	V-PR	V-ROC	Aff-F	V-PR	V-ROC	Aff-F	V-PR	V-ROC	Aff-F	V-PR	V-ROC
Qwen3-1.7B	72.33	48.53	<u>77.07</u>	65.22	45.83	71.96	65.21	39.69	<u>86.81</u>	<u>79.31</u>	55.53	93.29	<u>72.67</u>	48.52	<u>82.70</u>	69.25	59.14	81.65
Qwen2.5-1.5B	72.05	47.52	76.65	64.56	46.40	73.21	63.13	38.35	87.36	78.94	58.47	93.41	72.31	<u>50.83</u>	82.37	69.67	52.35	78.12
SmolLM2-1.7B	<u>72.13</u>	47.29	76.69	63.74	43.64	70.02	64.73	40.38	86.26	80.14	58.72	94.57	72.54	<u>45.80</u>	82.24	69.87	<u>60.35</u>	<u>82.22</u>
TinyLlama-1.1B	71.65	47.84	76.64	<u>65.60</u>	49.23	74.91	<u>65.05</u>	39.76	85.88	78.67	55.83	93.40	72.12	44.21	80.88	69.48	58.93	81.50
Phi-1.5	71.54	47.77	77.04	65.31	47.08	73.32	64.53	39.02	86.59	78.74	53.62	93.44	71.13	46.49	81.84	69.63	58.31	80.40
OLMo-1B	71.35	46.67	76.58	65.17	<u>47.32</u>	<u>74.61</u>	64.07	<u>40.47</u>	86.73	77.06	46.79	89.98	73.06	51.35	84.69	<u>69.76</u>	60.95	82.84
MobileLLaMA-1.4B	71.82	<u>48.27</u>	77.33	65.63	45.43	72.77	64.54	41.03	86.57	78.32	52.44	<u>93.56</u>	70.55	44.52	80.91	69.57	59.32	81.33

Table 3: Ablation results on six datasets. Aff-F uses the mean-threshold protocol; V-PR and V-ROC are threshold-independent. Values are percentages. Bold = best; underline = second best. –TB, –BP, –CS, and –SR remove the text branch, behavioral prompt, component semantics, and semantic reconstruction, respectively; T, S, and R denote trend, seasonal, and residual components.

Configuration	Weather			Energy			Env.			KR			EWJ			MDT		
	Aff-F	V-PR	V-ROC	Aff-F	V-PR	V-ROC	Aff-F	V-PR	V-ROC	Aff-F	V-PR	V-ROC	Aff-F	V-PR	V-ROC	Aff-F	V-PR	V-ROC
<i>Alignment objective</i>																		
$\lambda = 0$	71.69	46.03	76.47	64.71	46.26	72.53	62.57	31.35	81.58	78.13	51.26	91.47	72.58	45.04	78.98	66.16	44.54	72.58
Cos	71.59	46.62	76.63	62.82	41.25	69.12	64.30	39.36	85.84	78.26	53.62	91.92	72.24	43.30	80.55	66.16	46.24	73.75
MSE	71.75	46.19	76.25	66.33	<u>47.59</u>	<u>74.73</u>	64.73	38.97	85.67	<u>79.29</u>	56.07	92.59	72.49	46.87	82.70	66.94	47.05	74.62
Full	72.33	48.53	77.07	65.22	45.83	71.96	<u>65.21</u>	39.69	86.81	79.31	55.53	93.29	72.67	48.52	82.70	69.25	59.14	81.65
<i>Text and reconstruction paths</i>																		
–TB	69.62	47.38	76.14	63.82	42.53	71.31	61.28	30.04	77.52	72.85	44.22	83.82	64.35	30.11	71.66	66.34	44.60	72.11
–BP	72.71	52.64	79.92	65.01	47.73	74.92	66.48	35.53	82.61	73.96	38.08	85.37	65.14	27.76	64.77	64.51	32.79	64.60
–CS	71.47	47.02	76.53	64.51	44.97	71.08	63.45	38.93	85.52	77.91	47.89	90.52	71.46	47.40	80.79	66.96	43.63	72.04
–SR	70.37	48.07	76.21	64.19	43.24	71.43	63.17	34.07	81.31	77.22	46.39	83.48	68.05	34.77	71.98	63.50	36.99	65.38
<i>Component subset</i>																		
T	71.24	48.08	76.75	66.57	47.04	74.43	63.47	<u>39.56</u>	86.94	77.76	54.03	93.42	72.20	47.18	83.92	67.87	55.23	<u>79.51</u>
S	71.09	47.65	76.53	66.37	46.70	74.07	64.41	38.68	85.89	77.77	53.12	93.56	72.21	47.16	83.70	67.87	55.39	79.30
R	71.13	47.50	76.34	<u>66.62</u>	47.41	73.92	63.59	38.02	85.54	77.53	53.06	<u>93.54</u>	72.18	47.62	84.50	68.18	56.01	79.21
T+S	71.77	49.55	77.88	65.55	44.52	71.80	64.44	39.51	<u>86.82</u>	79.16	58.27	93.20	<u>72.71</u>	48.73	<u>84.57</u>	68.58	56.25	78.88
T+R	72.25	<u>49.70</u>	<u>78.05</u>	66.00	45.53	72.27	63.52	38.14	85.53	78.90	<u>56.98</u>	92.39	72.95	<u>48.69</u>	84.67	<u>68.87</u>	<u>56.82</u>	78.97
S+R	71.70	49.05	77.59	67.03	45.52	72.50	64.05	38.87	85.62	78.44	56.07	92.31	72.42	46.95	83.09	68.64	56.73	78.77

Across the seven backbones, average Aff-F ranges from 70.07% to 70.67% (0.60 points), V-PR from 48.50% to 49.54% (1.04 points), and V-ROC from 81.85% to 82.57% (0.72 points). Dataset-level winners also vary: for example, SmolLM2-1.7B is strongest in Aff-F on KR and MDT, whereas OLMo-1B is strongest on EWJ. The small aggregate ranges indicate limited sensitivity to the choice among the evaluated lightweight LLM backbones. They do not establish insensitivity to larger or differently pretrained language models, nor do they identify which framework module is responsible for the observed stability.

Ablation Study

Table 3 reports results for all six datasets; the comparisons below use the corresponding six-dataset averages. *No alignment* ($\lambda = 0$) retains the text and reconstruction paths. Cos and MSE replace Gaussian NLL with cosine distance and deterministic squared error. –TB deletes both text encoders, alignment, and text-conditioned reconstruction, so masked temporal features are reconstructed directly. –BP removes the behavioral-prompt encoder and prompt-component cross-attention, using fused component semantics as **Z**. –CS removes decomposition, role-prompt encoders, and component fusion, using only the behavioral-prompt representation. –SR retains text construction and alignment but bypasses text-conditioned reconstruction. The subset variants retain only the named branches and zero-fill the remainder before the shared fusion projection. Because

Figures/hyper.pdf

Figure 2: Sensitivity of Aff-F and V-PR to the mask ratio and alignment-loss weight λ on Environment and Energy. Vertical dashed lines indicate the default settings.

the variants remove different paths, their values should not be

Figures/text_to_time.pdf

Figure 3: Patch-level text-to-time Gaussian alignment on a representative EWJ segment. The left panel shows the raw alignment NLL, its smoothed trend, the predicted standard deviation σ , and ground-truth anomalous intervals. The right panel compares the NLL distributions of normal and anomalous patches.

read as a strict ordering of isolated component contributions.

The full Gaussian-NLL configuration improves over no alignment by 1.36 points in Aff-F, 5.46 in V-PR, and 3.32 in V-ROC. It also exceeds deterministic MSE alignment by 0.41, 2.42, and 1.16 points, respectively. These comparisons suggest that predicting both the mean and uncertainty of temporal patch representations is more effective than deterministic matching under this protocol. Removing the text branch decreases the three averages by 4.29, 9.73, and 6.82 points; removing semantic reconstruction decreases them by 2.92, 8.95, and 7.28 points. Removing component semantics yields drops of 1.37, 4.57, and 2.84 points, while removing the behavioral prompt yields drops of 2.70, 10.45, and 6.88 points. These results indicate that both textual paths and their use in reconstruction are associated with the aggregate gains.

Among component subsets, the two-component variants approach the full model in Aff-F and V-PR, but none exceeds it on both metrics. Trend-only obtains the highest V-ROC (82.50), only 0.25 points above the full model. This small exception suggests that the preferred component mixture depends on the metric, whereas the full configuration provides the strongest overall trade-off.

Hyperparameter Sensitivity Analysis

Figure 2 shows dataset-dependent responses to the mask ratio. On Environment, Aff-F reaches a relatively high level around ratios of 0.4–0.5. Energy remains comparatively stable from 0.1 to 0.5 but declines markedly at 0.6. The results indicate that moderate masking provides a robust operating

range, whereas excessive masking can impair reconstruction quality, as is particularly evident on Energy. At lower ratios, the reconstruction task may become less challenging because more temporal context remains visible, although the figure does not imply a uniform degradation across datasets.

For the alignment-loss weight, Aff-F on both Environment and Energy improves overall as λ increases from 0 to 1, while $\lambda = 2$ yields no further improvement. This behavior provides evidence that the Gaussian alignment objective is beneficial, while also indicating that its contribution should remain balanced with the reconstruction objective rather than dominate optimization.

Visualization and Case Study

Figure 3 examines the internal text-to-time alignment behavior. The temporal NLL curve contains pronounced local fluctuations, so it should not be interpreted as a point-wise anomaly detector. More reliable evidence is provided by the distributional comparison: anomalous patches exhibit a higher NLL median and a wider distribution than normal patches, indicating a stronger representation mismatch between these patches and the Gaussian distribution predicted from text. This diagnostic is consistent with the intended role of probabilistic alignment. Alignment NLL is used only to inspect this internal behavior; the final anomaly score is still derived from reconstruction error.

Figure 4 shows distinct successes and limitations across the six datasets. On KR, several short input spikes produce sharp score peaks, including the early and late annotated events, but a large response near the middle of the plot falls outside the nearest shaded interval and is a likely false positive. EWJ contains a longer, highly variable central regime: the detector sustains many above-threshold responses within it, yet the fragmented peaks blur its boundaries and some responses extend beyond annotated regions. On MDT, the isolated spikes near the middle and toward the end are localized sharply, whereas the broader central anomalous interval elicits only intermittent responses and includes a few off-interval peaks. Environment exhibits strong responses around the dense abnormal activity near the center and right, but the elevated score spreads across interval boundaries, limiting precise onset and offset localization. On Energy, the detector produces a sustained cluster of high scores through the main long anomaly, while its response is weaker near the beginning and end of the shaded ranges and does not cover the entire duration uniformly. Finally, Weather captures the prominent short event near the center and the late anomalous spikes, but several threshold crossings in nominal regions indicate possible false alarms. The cases therefore support sensitivity to both short spikes and sustained deviations, while also exposing boundary imprecision, incomplete coverage of long events, and occasional responses outside annotated intervals.

Conclusion

Figures/detection_case.pdf

Figure 4: Qualitative anomaly-score examples on six datasets. For each dataset, the top row shows the input series with ground-truth anomalous intervals, and the bottom row shows the reconstruction-based anomaly scores and detection thresholds.